

Module 1: Number Systems — Summary

What this module is about. Numbers can be organized by the kinds of values they allow, but the same number can also be viewed in different ways depending on what we want to do with it. Addition is naturally interpreted as motion on a number line. Multiplication is naturally interpreted as scaling. Fractions can look different while representing the same location. Recursive rules describe how a quantity changes one step at a time.

The standard number systems are nested:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

- $\mathbb{N}$: natural numbers $1, 2, 3, \dots$. In this course, $0 \notin \mathbb{N}$, so 0 is first classified as an integer.
- $\mathbb{Z}$: integers such as $-4, 0, 12$.
- $\mathbb{Q}$: rational numbers, which can be written as a/b with integers a, b and $b \neq 0$.
- $\mathbb{R}$: all real numbers, including irrational numbers such as $\sqrt{2}$ and π .

When asked for the *smallest* standard system containing a number, choose the first set in this chain that contains it.

Example 1: Signed numbers record direction

A diver begins at the surface. The diver descends 15 meters, rises 8 meters, and then descends another 5 meters. Let upward motion be positive and downward motion be negative.

$$0 + (-15) + 8 + (-5) = -12.$$

The diver finishes 12 meters below the surface. Once direction has been encoded by signs, the entire trip can be written using addition.

- Adding c moves a number by a fixed amount: $x \mapsto x + c$.
- Multiplying by c rescales a number: $x \mapsto cx$.
- If $0 < c < 1$, multiplication by c shrinks a positive number toward 0.
- Two fractions a/b and c/d are equivalent when $ad = bc$.

For example,

$$\frac{12}{14} = \frac{6}{7}, \quad \frac{18}{21} = \frac{6}{7},$$

so the two fractions represent the same rational number. The cross-products also agree: $12 \cdot 21 = 14 \cdot 18 = 252$.

An **additive recursion** changes by the same amount every step:

$$a_{n+1} = a_n + d, \quad a_n = a_0 + nd.$$

A **multiplicative recursion** changes by the same factor every step:

$$b_{n+1} = rb_n, \quad b_n = b_0 r^n.$$

A constant difference signals additive change. A constant ratio signals multiplicative change.

Example 2: Fixed addition versus percentage growth

Two populations begin at 1000. Population A gains 200 people each year. Population B grows by 15% each year.

$$A_n = 1000 + 200n, \quad B_n = 1000(1.15)^n.$$

After one year, $A_1 = 1200$ while $B_1 = 1150$, so the fixed-addition model is initially larger. But Population B's yearly increase is itself growing because 15% is applied to the current population. This is why percentage growth can eventually overtake a larger-looking fixed addition.

By the end of this module, you should be able to:

- classify a number using the smallest standard number system that contains it;
- use signed numbers to represent direction and position;
- recognize multiplication as scaling and test whether two fractions are equivalent;
- generate terms of additive and multiplicative recursions and distinguish constant difference from constant ratio.

Module 1: Number Systems — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Useful facts

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, additive recursion: $a_{n+1} = a_n + d$, multiplicative recursion: $b_{n+1} = rb_n$.

1. A checking account is overdrawn by \$35, so its balance is represented by -35 . Using the smallest standard number system that contains the value, classify -35 as natural, integer, rational, or real but irrational. Briefly state why your choice is the smallest one that works.

2. A hiker is at position -6 on a trail map. The hiker walks 10 units forward and then 14 units backward. Treat forward motion as positive and backward motion as negative.
 - (a) Write the entire trip as a single addition expression using signed numbers.
 - (b) Find the hiker's final position.

3. A recipe uses the fraction $6/9$, while another version of the same recipe uses $14/21$.
 - (a) Use cross-products to decide whether the fractions are equivalent.
 - (b) Simplify both fractions to confirm your conclusion.

4. A neighborhood has 800 households at the beginning of year 0. A planning model assumes that the neighborhood gains exactly 120 households each year.
 - (a) Write an additive recursion for the number of households H_n .
 - (b) Find H_4 , the number of households at the beginning of year 4.

5. A second neighborhood also begins with 800 households, but it grows by 10% each year.

- (a) Write the multiplicative recursion.
- (b) Find the population after 3 years.
- (c) Which feature of the model is constant from year to year: the dollar-size increase, the difference, or the ratio?

Module 1: Number Systems — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

1. A temperature is recorded as -8°C . Using the smallest standard number system that contains the numerical value -8 , classify it as natural, integer, rational, or real but irrational.
2. An elevator begins on floor -3 , rises 11 floors, and then descends 6 floors. Write the trip as an addition expression with signed numbers and determine the final floor.
3. Decide whether $8/12$ and $18/27$ represent the same rational number. Show either a cross-product check or a simplification of both fractions.
4. A town has 1200 homes and adds 75 homes each year. Write an additive recursion and find the number of homes after 5 years.
5. A different town begins with 1200 homes and grows by 5% per year. Write a multiplicative recursion and find the number of homes after 2 years.